Muhammet Ozdemir

AI Engineer & Researcher

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SUMMARY

Lead developer on TUBITAK and TUSEB-funded AI and computer-vision research projects; developer of novel scientific methods in AutoML and AI contributing to the literature.

CORE SKILLS

AI / ML AutoML, AI Optimization, Computer Vision, Machine Learning, Deep Learning, MLOps

Applied Image Processing, Data Analysis, Model Deployment, Research & Publication

Tools Python, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, SQL, CUDA, Docker, Git

EXPERIENCE

AI Engineer · *Turkiye* Sept 2025 – Present

- Built face-verification systems with a 1-in-10M error rate, plus ID recognition, OCR, and liveness modules for banking applications; owned the full TSE certification process.

AI Researcher (Full-time) — Erciyes University AI & Big Data Research Center · *Remote* Oct 2024 – Present

- Conduct AutoML and AI-optimization research on activation-function design, Neural Architecture Search (NAS), and end-to-end AutoML design, published in SCI/E-indexed venues.

AI Engineer Intern — Skechers Turkiye / Olka Jun – Aug 2026

- On-premises LLM serving with vLLM, Text-to-SQL, a centralized RAG architecture, and CNN-based document processing.
- Dockerized services, integration with internal applications, and KVKK-compliant guardrails.

PROJECTS

ABC-Tuned ACO Fleet Optimization Mar 2026 – Present

- Developing a hybrid fleet-optimization system combining demand forecasting, Artificial Bee Colony hyperparameter tuning, and Ant Colony Optimization for cost-aware shipment planning.

Hybrid Mammography Analysis — Early Breast-Cancer Detection 2024

- Built hybrid CNN architectures on 4,000+ mammograms for lesion detection; achieved >90% BI-RADS accuracy. National finalist (presented in Antalya). Funded by TUSEB & Ministry of Health.

ML-Based Customer Matching & Product Recommendation 2023

- Built a hybrid ML engine combining clustering with user/item collaborative filtering; reached >90% match accuracy and integrated it into POS and mobile apps. Funded by TUBITAK.

SCIENTIFIC PUBLICATIONS

- Automatic Design of Deep Neural Network Activation Functions Using Genetic Programming. *SCI/E-indexed (Q3) — revision submitted*
- Analysis of Evolutionary and Genetic Programming Mutation Operators for Automated Activation Function Design in Deep Neural Networks. *SCI/E-indexed (Q2) — revision submitted*

EDUCATION & CERTIFICATIONS

B.S. in Computer Engineering — Erciyes University, Kayseri, Turkiye Sept 2021 – Graduated July 2026

English Language, B1+ – B2 — St Giles International, United Kingdom Sept 2025

LEADERSHIP

Founder & President — ERU AI Club Mar 2024 – Present

- Led 20 CS students across 8 TUBITAK-funded AI projects and a TUSEB-backed health-AI initiative (national finalist); co-authored the resulting publications.